VSDBABYSOC

POST-ROUTE MULTI-CORNER STATIC TIMING ANALYSIS

Sky130HD • 16 PVT Corners • Setup and Hold Assessment

NAME	T.MURALI
ROLLNO	24EG104C35
SECTION	ECE-C
DESIGN	VSDBabySOC
TECHNOLOGY	OpenROAD, OpenSTA, Sky130HD PDK, Tcl, Excel.
Analaysis	Post-Route Multi-Corner Static Timing Analysis (STA) of VSDBabySoC across 16 Sky130HD PVT corners, evaluating setup/hold slack, WNS, and TNS.

1. Objective:

This report documents post-route static timing analysis (STA) of the VSDBabySoC design across 16 Sky130HD process-voltage-temperature (PVT) library corners. It explains the six timing metrics shown in the Excel sheets, summarizes the measured values, includes the supplied charts and OpenROAD timing screenshot, and records the checkpoints required to reproduce the analysis.

2. Design and Analysis Setup:

Item	Description
Design	VSDBabySoC
Technology	SkyWater SKY130HD standard-cell library
Analysis stage	Post-route STA
Physical database	5_route.odb / 6_final.odb, according to the loaded workflow stage
Timing constraints	Routed-stage SDC; the workflow used 6_1_fill.sdc and also inspected updated_clks.sdc
Parasitics	6_final.spef
Tool	OpenROAD (GUI and Tcl command interface)
Corner count	16 standard-cell Liberty PVT corners

3. STA Flow and Reproducibility Checkpoints:

- Confirm the flow root, design directory, poststa directory, platform files, and result files exist.
- Confirm the 16 Sky130HD standard-cell Liberty files are available and each filename maps to the intended PVT corner.
- Load the required technology and macro LEF/Liberty data, then read the routed OpenDB database.
- Read the SDC that belongs to the selected database stage and verify the clock definition and exceptions.
- Load the extracted routed parasitics from 6_final.spef after setting the platform RC model.
- Run setup and hold reports for each corner; save the raw reports and capture the six summary metrics.
- Export the values to Excel, use the PVT corner labels as categories, and plot each metric separately.
- Review check_setup output for missing input/output delays and unconstrained endpoints before making sign-off claims.

4. Meaning of the PVT Corner Names:

PVT means Process, Voltage, and Temperature. The prefix ff denotes a fast-fast process model, ss denotes a slow-slow model, and tt denotes a typical-typical model. The temperature and voltage labels are embedded in each filename. The exact voltage and temperature definitions should be verified from the header of the corresponding Liberty file; filenames alone are not sufficient for sign-off.

Group	General interpretation
FF corners	Fast-process models; often useful for checking short-path/hold behavior.
SS corners	Slow-process models; often critical for setup timing.
TT corners	Typical-process reference models.
Voltage/temperature suffixes	Operating conditions encoded in the library filename; verify in the Liberty header.

5. Timing Metrics Explained:

Metric	Meaning	How to interpret
Setup WS (Worst Slack)	Worst setup slack among the analyzed setup paths.	Positive: reported path meets setup; negative: setup violation.
Setup WNS	Worst negative setup slack as summarized by the reporting script.	A negative value indicates the worst setup violation; zero can mean no negative slack was captured.
Setup TNS	Sum of negative setup slack across violating endpoints.	Negative/non-zero magnitude indicates aggregate setup violations; zero does not prove all paths are constrained.
Hold WS (Worst Slack)	Worst hold slack among analyzed hold paths.	Positive: reported path meets hold; negative: hold violation.
Hold WNS	Worst negative hold slack.	Zero means the report captured no negative hold slack.
Hold TNS	Total negative hold slack.	Zero means no negative hold slack was accumulated by the report.

6. Setup WS — Results and Interpretation:

Sky130HD PVT Corner	Setup WS (ns)
ff_100C_1v65	8.20
ff_100C_1v95	9.06
ff_n40C_1v56	7.05
ff_n40C_1v65	7.76
ff_n40C_1v76	8.38
ff_n40C_1v95	9.07
ss_100C_1v40	-3.31
ss_100C_1v60	1.75
ss_n40C_1v28	-28.66
ss_n40C_1v35	-15.83
ss_n40C_1v40	-10.32
ss_n40C_1v44	-7.16
ss_n40C_1v60	0.20
ss_n40C_1v76	3.62
tt_025C_1v80	7.03
tt_100C_1v80	7.11

Interpretation: the most negative setup WS in the supplied spreadsheet is −28.66 ns at ss_n40C_1v28. Other negative entries are ss_100C_1v40 (−3.31 ns), ss_n40C_1v35 (−15.83 ns), ss_n40C_1v40 (−10.32 ns), and ss_n40C_1v44 (−7.16 ns). These SS corners require investigation. The FF and TT entries and the remaining listed SS entries have positive setup WS in this supplied dataset.

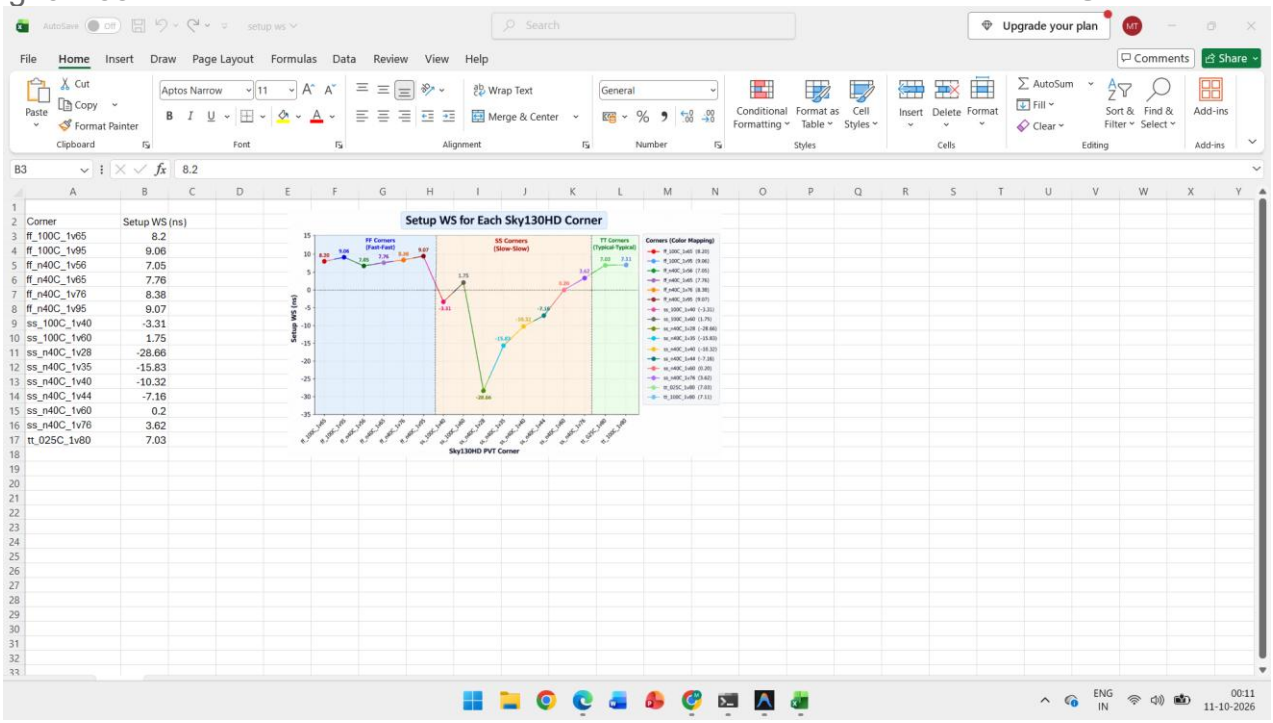
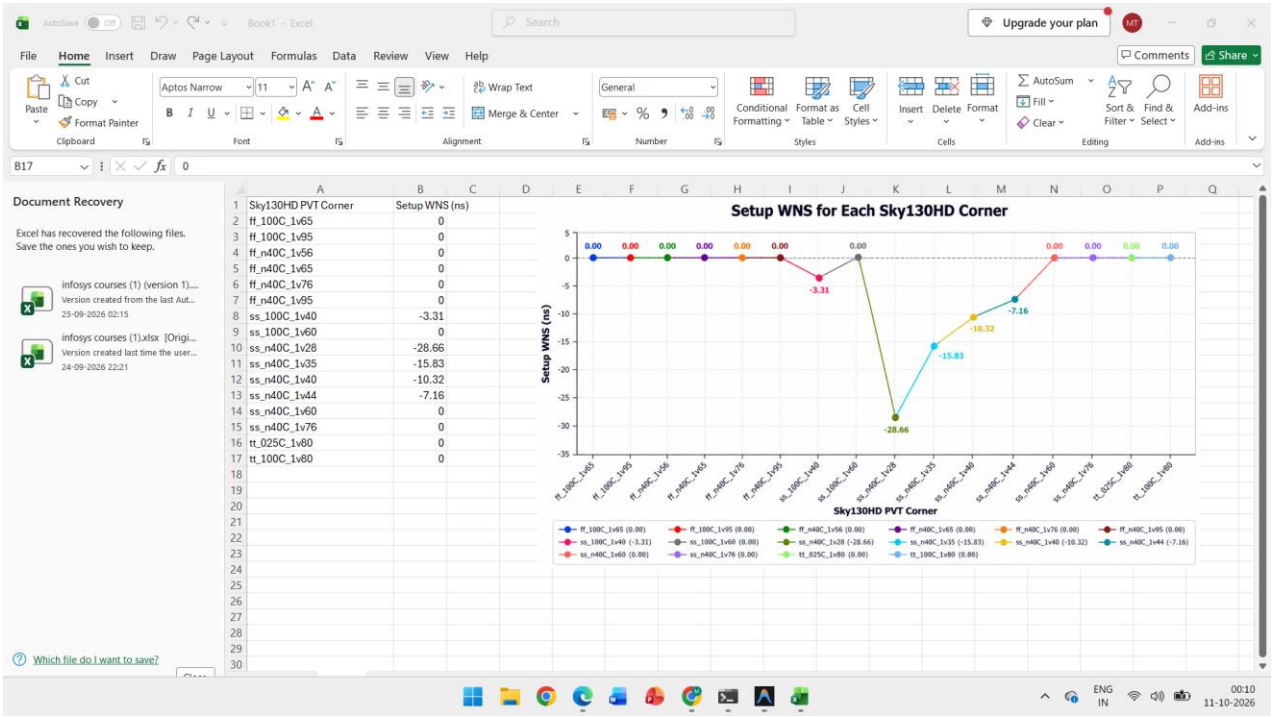


Figure 1. Setup WS by Sky130HD PVT corner (Excel chart supplied by the user).

7. Setup WNS — Results and Interpretation:

The supplied Setup WNS chart reports zero for most corners, with negative values in the SS region. The most negative plotted value is approximately −28.66 ns at ss_n40C_1v28; other visible violations include approximately −15.83 ns at ss_n40C_1v35, −10.32 ns at ss_n40C_1v40, −7.16 ns at ss_n40C_1v44, and −3.31 ns at ss_100C_1v40. These values identify corners that need path-level review.



8. Setup TNS — Results and Interpretation:

Setup TNS accumulates negative setup slack across violating endpoints. The supplied chart shows large negative totals for selected SS corners, with the largest magnitude near −13,598.72 ns at ss_n40C_1v28. Other plotted negative totals include approximately −5,921.85 ns at ss_n40C_1v35, −2,880.22 ns at ss_n40C_1v40, −1,573.23 ns at ss_n40C_1v44, and −188.34 ns at ss_100C_1v40. Treat these as the values displayed in the supplied Excel evidence and confirm them against raw per-corner reports.

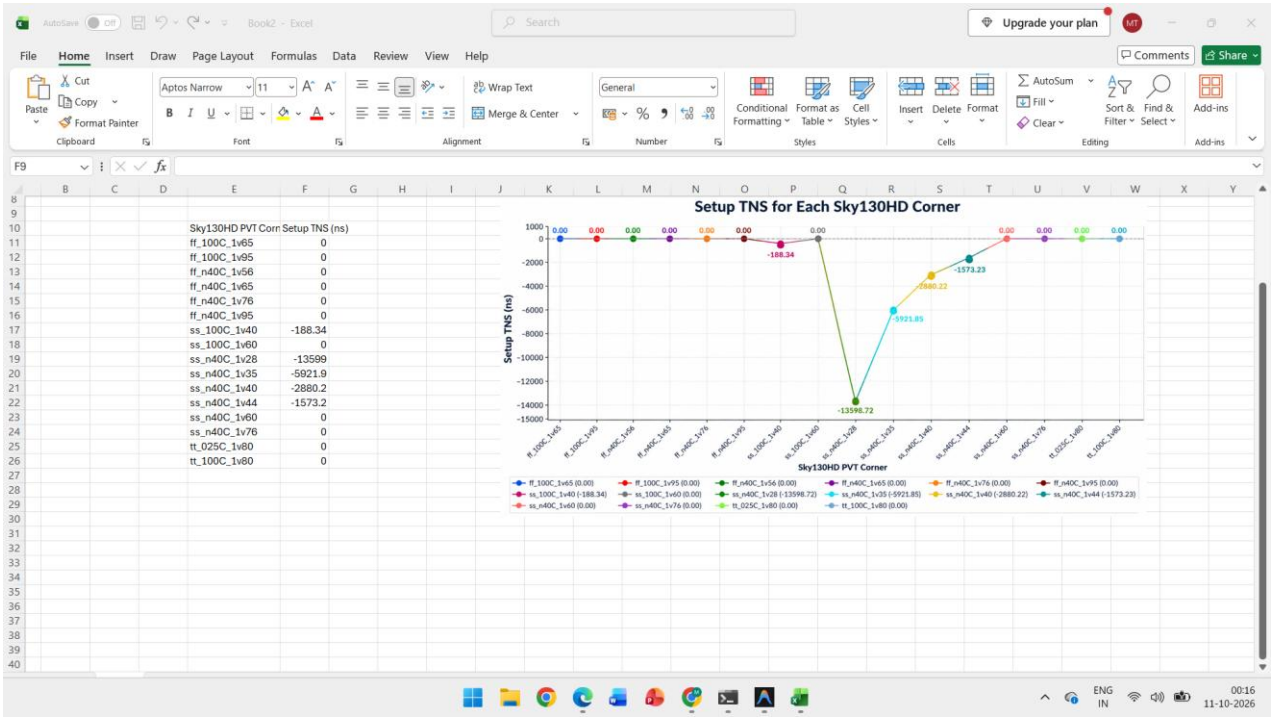


Figure 3. Setup TNS by PVT corner (Excel chart supplied by the user).

9. Hold WS — Results and Interpretation:

Sky130HD PVT Corner	Hold WS (ns)
ff_100C_1v65	0.26
ff_100C_1v95	0.20
ff_n40C_1v56	0.29
ff_n40C_1v65	0.27
ff_n40C_1v76	0.24
ff_n40C_1v95	0.20
ss_100C_1v40	0.92
ss_100C_1v60	0.65
ss_n40C_1v28	1.84
ss_n40C_1v35	1.36
ss_n40C_1v40	1.14
ss_n40C_1v44	1.00
ss_n40C_1v60	0.67
ss_n40C_1v76	0.50
tt_025C_1v80	0.32
tt_100C_1v80	0.33

Interpretation: all 16 supplied Hold WS values are positive. The smallest displayed value is 0.20 ns at ff_100C_1v95 and ff_n40C_1v95; the largest is 1.84 ns at ss_n40C_1v28. This suggests that the reported hold paths meet hold timing in the supplied comparison, subject to the constraint and model caveats.

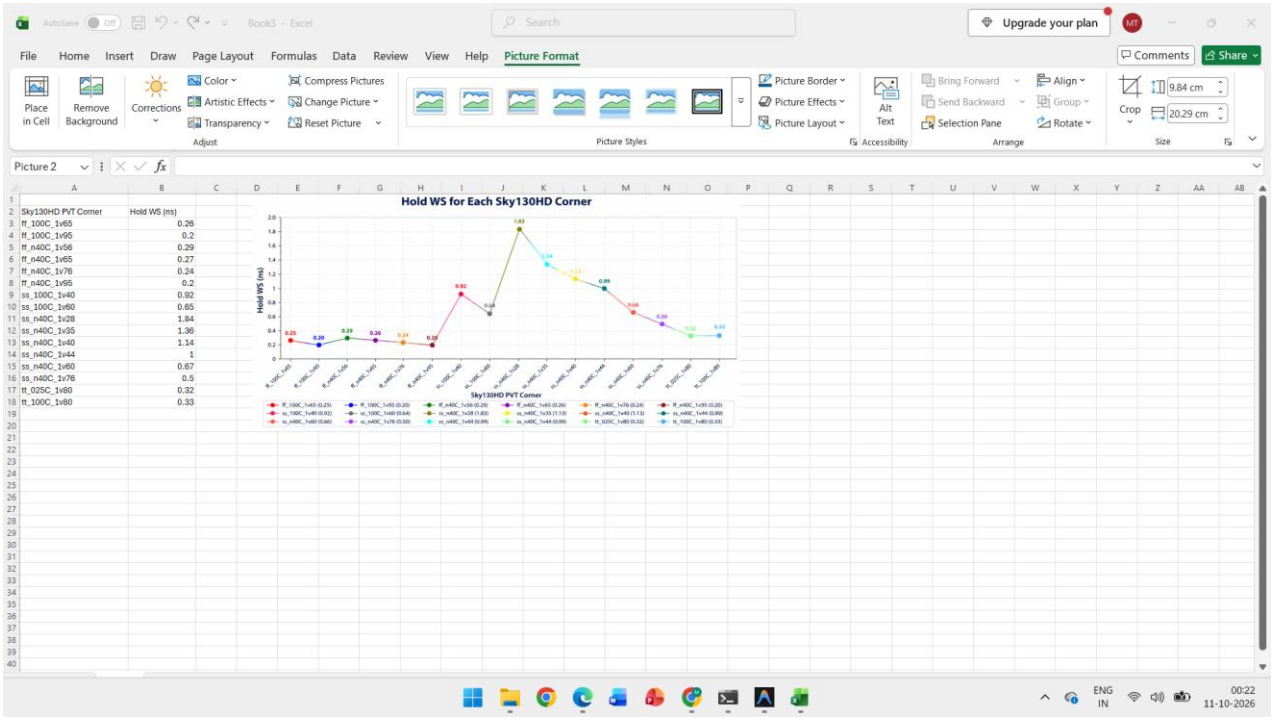


Figure 4. Hold WS by PVT corner (Excel chart supplied by the user).

10. Hold WNS — Results and Interpretation:

The supplied Hold WNS chart is flat at zero across all 16 corners. In this dataset, no negative hold slack was captured by the summary report. A zero WNS result should still be checked against the raw per-corner timing report and the set of constrained endpoints.

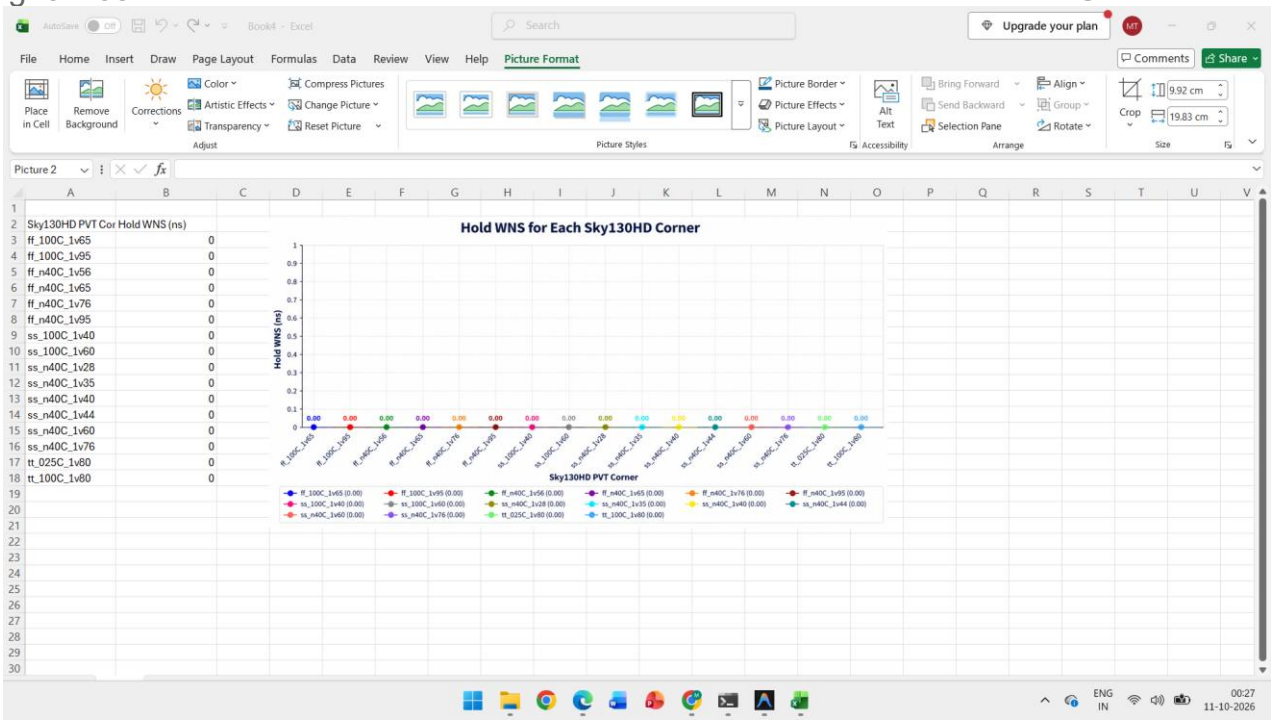


Figure 5. Hold WNS by PVT corner (Excel chart supplied by the user).

11. Hold TNS — Results and Interpretation:

The supplied Hold TNS chart is also flat at zero for every corner, meaning the summary reports no accumulated negative hold slack. This is consistent with the positive Hold WS values shown in Figure 4, but it does not independently establish sign-off completeness.

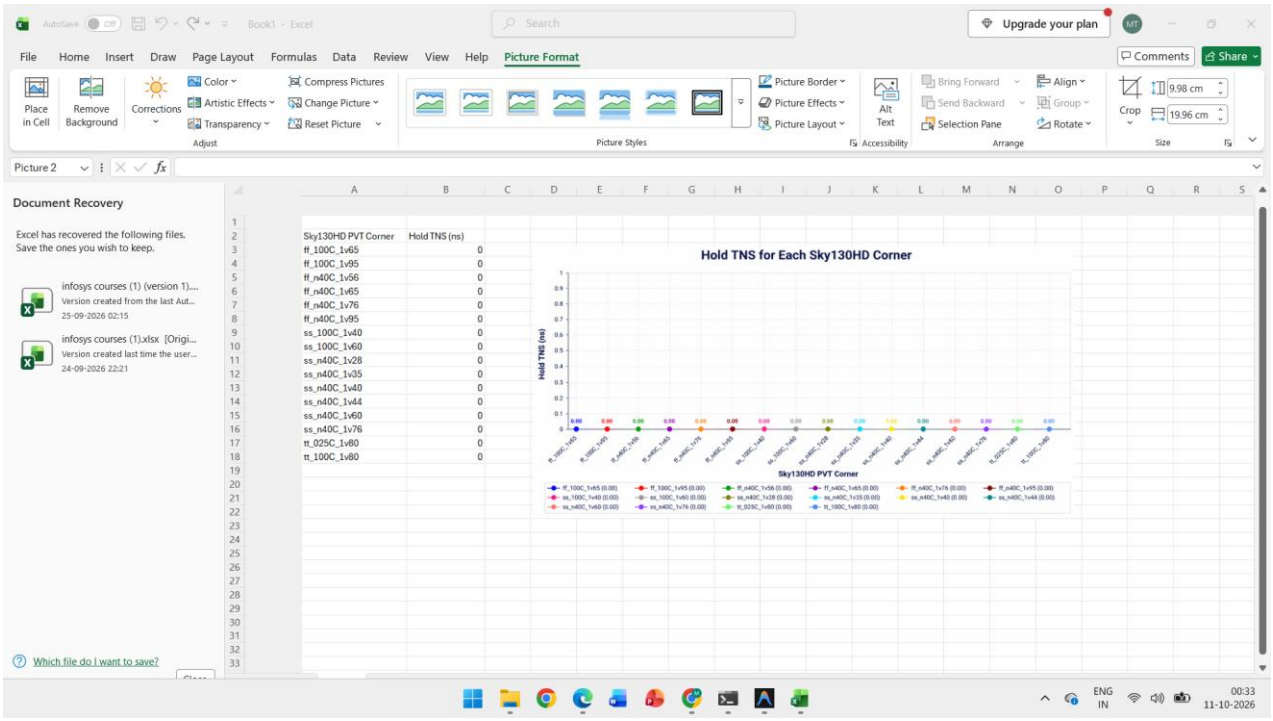


Figure 6. Hold TNS by PVT corner (Excel chart supplied by the user).

12. OpenROAD Post-Route Timing Screenshot:

The supplied OpenROAD GUI screenshot shows a timing path report with data required time of 12.80 ns, data arrival time of −5.97 ns, and slack of 6.83 ns (MET) for the displayed path. This is an example path-level report from the loaded routed design. It should be kept distinct from the per-corner Excel summary, because the exact library corner, SDC, and database state used for that GUI screenshot must be recorded.

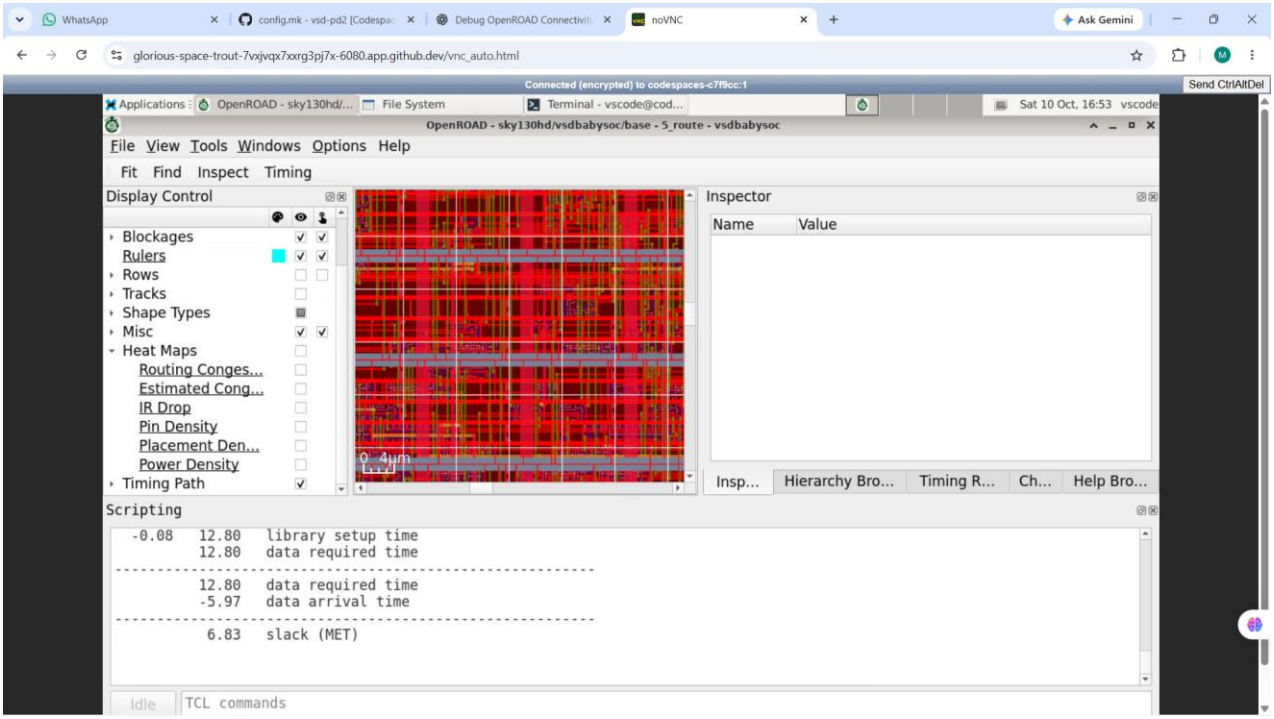


Figure 7. OpenROAD GUI / timing evidence supplied during the post-route STA workflow.

13. Overall Findings:

- Setup timing is the main concern in the supplied 16-corner comparison, particularly at ss_n40C_1v28.
- The other negative setup WS corners are ss_100C_1v40, ss_n40C_1v35, ss_n40C_1v40, and ss_n40C_1v44.
- All 16 Hold WS values shown in the supplied table are positive; Hold WNS and Hold TNS charts show zero for all corners.
- FF and TT corners show positive Setup WS in the supplied values, while several SS corners show setup violations.
- The charts summarize the supplied spreadsheet evidence; raw STA reports should be retained to confirm exact values and corner-to-report mapping.

14. Important Limitations and Sign-Off Checks:

The earlier check_setup output identified missing input delays for ENb_CP, ENb_VCO, REF, VCO_IN, VREFH, and reset; a missing output delay for OUT; and unconstrained endpoints including OUT and core.CPU_reset_a1$_DFF_P_/D. These must be addressed or explicitly justified before claiming timing

sign-off. Do not invent I/O delays: obtain interface timing requirements from the design specification or project owner.

- Verify that the selected SDC corresponds to the exact routed database stage being analyzed.
- Confirm all intended clocks are defined and propagated as appropriate for the chosen stage.
- Confirm every Liberty file is actually loaded and associated with the correct corner.
- Confirm macro timing models and corner coverage; a fixed macro model across standard-cell corners can limit corner accuracy.
- Review check_setup -verbose and reports of unconstrained paths before interpreting zero WNS/TNS as proof of closure.
- Re-run setup and hold reports per corner after correcting constraints and any model-loading issues.

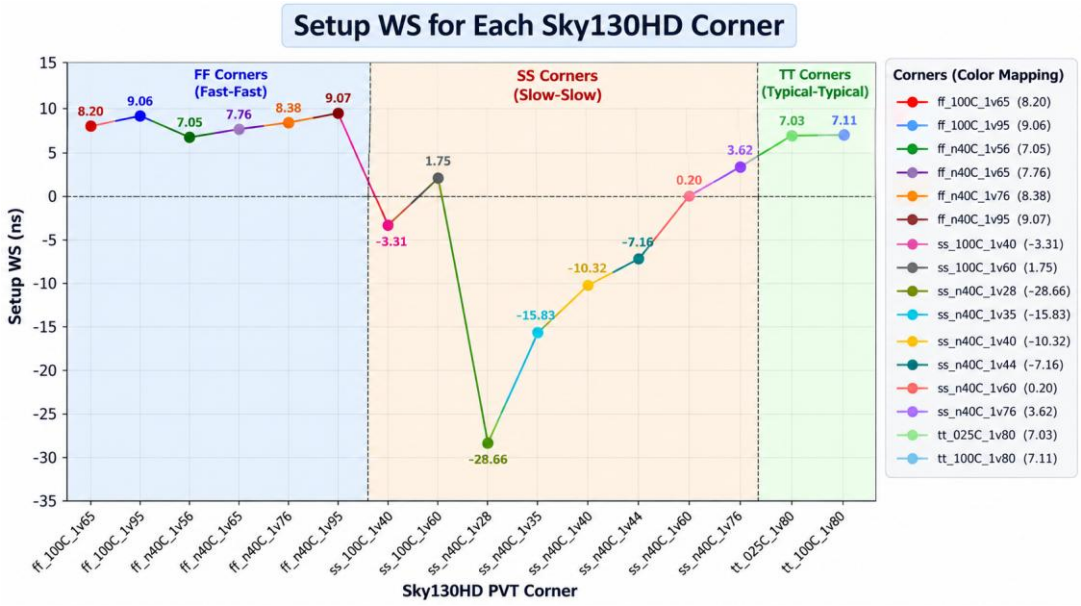
15. Suggested Evidence and File Organization:

Folder / file	Purpose
poststa/sta.tcl	Repeatable driver script for per-corner analysis.
poststa/reports/summary.csv	One row per corner with the six timing metrics.
poststa/reports/<corner>_setup.rpt	Detailed setup timing report for each corner.
poststa/reports/<corner>_hold.rpt	Detailed hold timing report for each corner.
poststa/reports/check_setup.rpt	Constraint coverage and unconstrained endpoint review.
Excel workbook	Source values and six charts, with consistent corner ordering and units.

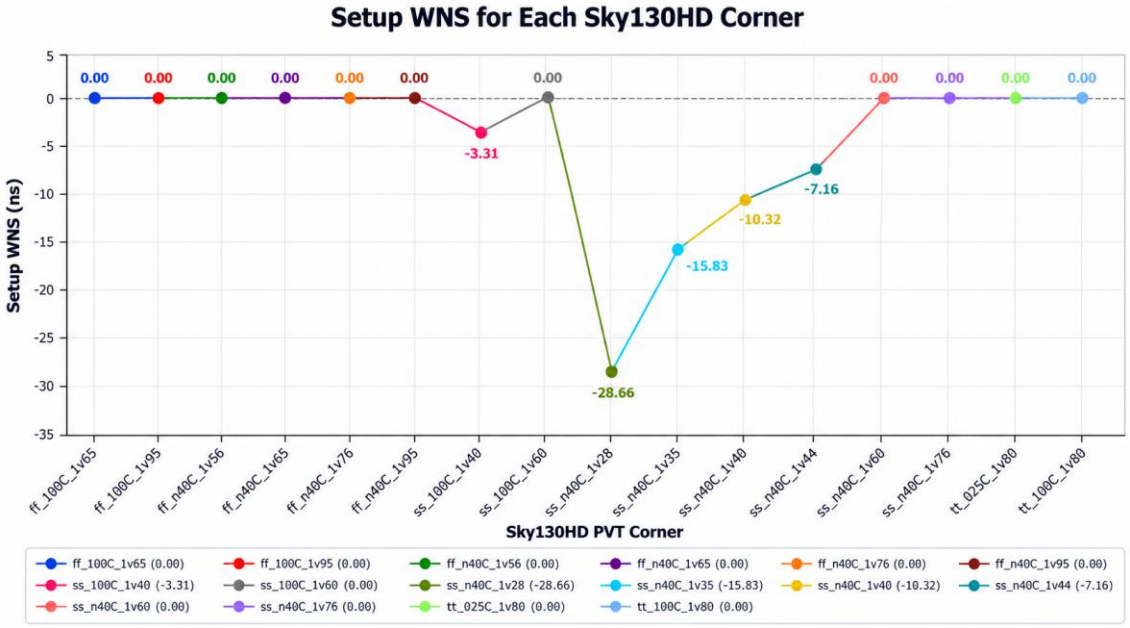
16. Conclusion:

The post-route multi-corner STA evidence indicates that setup timing is not met at several SS PVT corners, with ss_n40C_1v28 showing the worst supplied Setup WS of −28.66 ns. The supplied hold results are positive for all corners, and Hold WNS/TNS remain zero. The next engineering step is to investigate the worst SS setup paths and verify complete timing constraints and corner/model consistency. Until missing I/O delays and unconstrained endpoints are resolved, this report should be treated as an STA assessment rather than a final sign-off report.

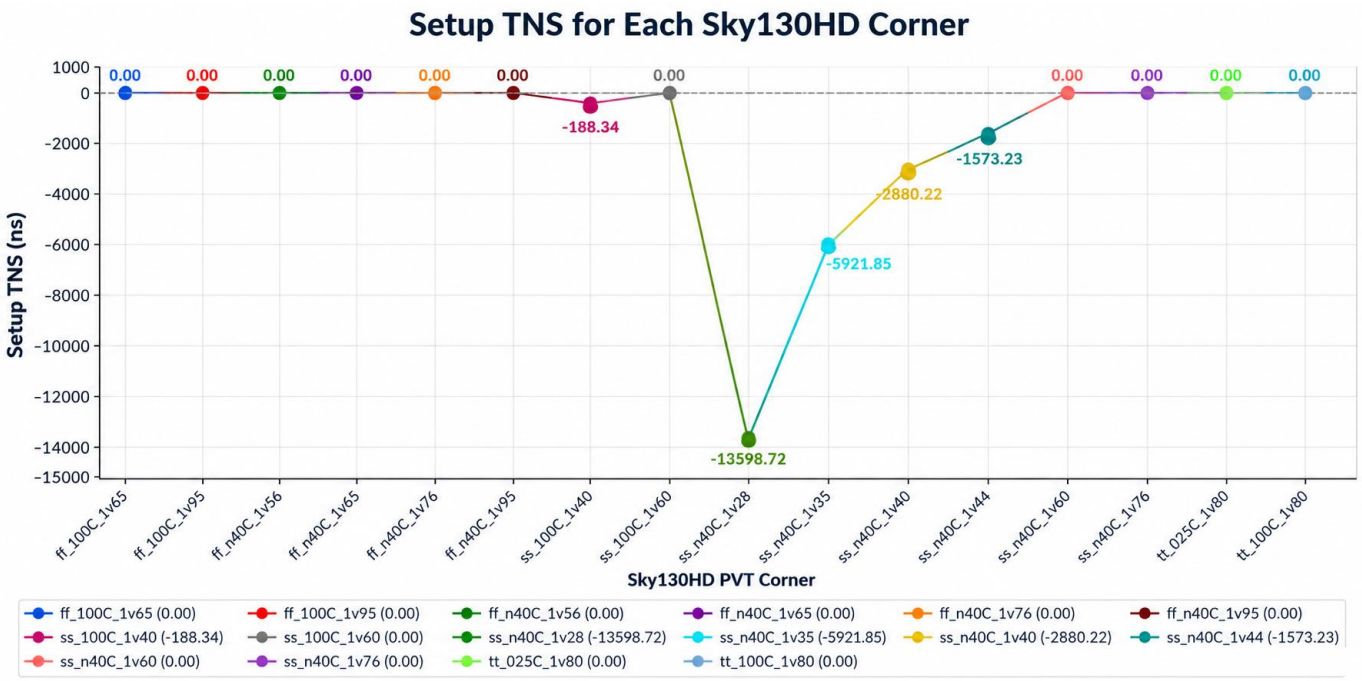
APPENDIX A:SETUP WS



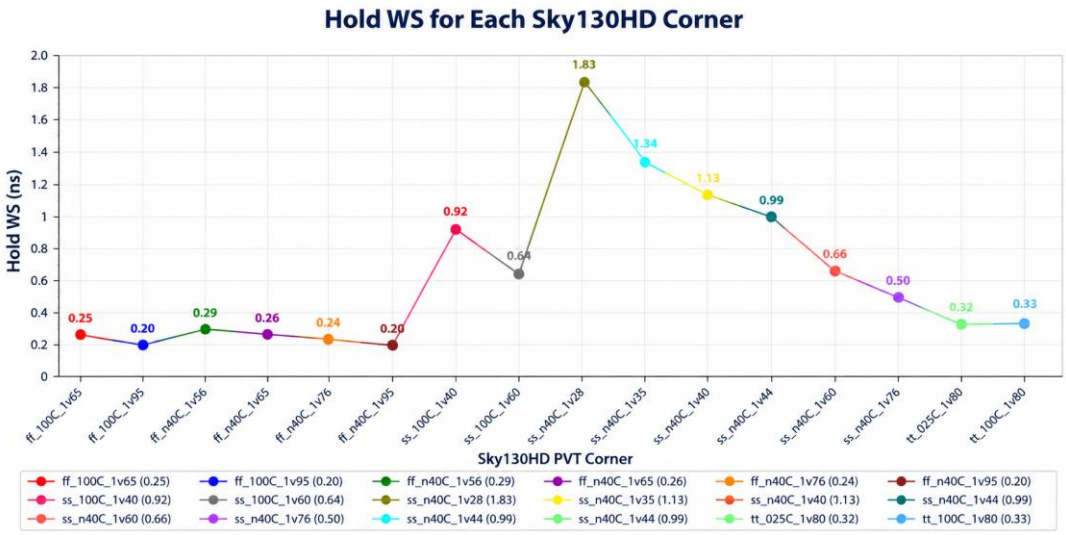
APPENDIX B:SETUP WNS



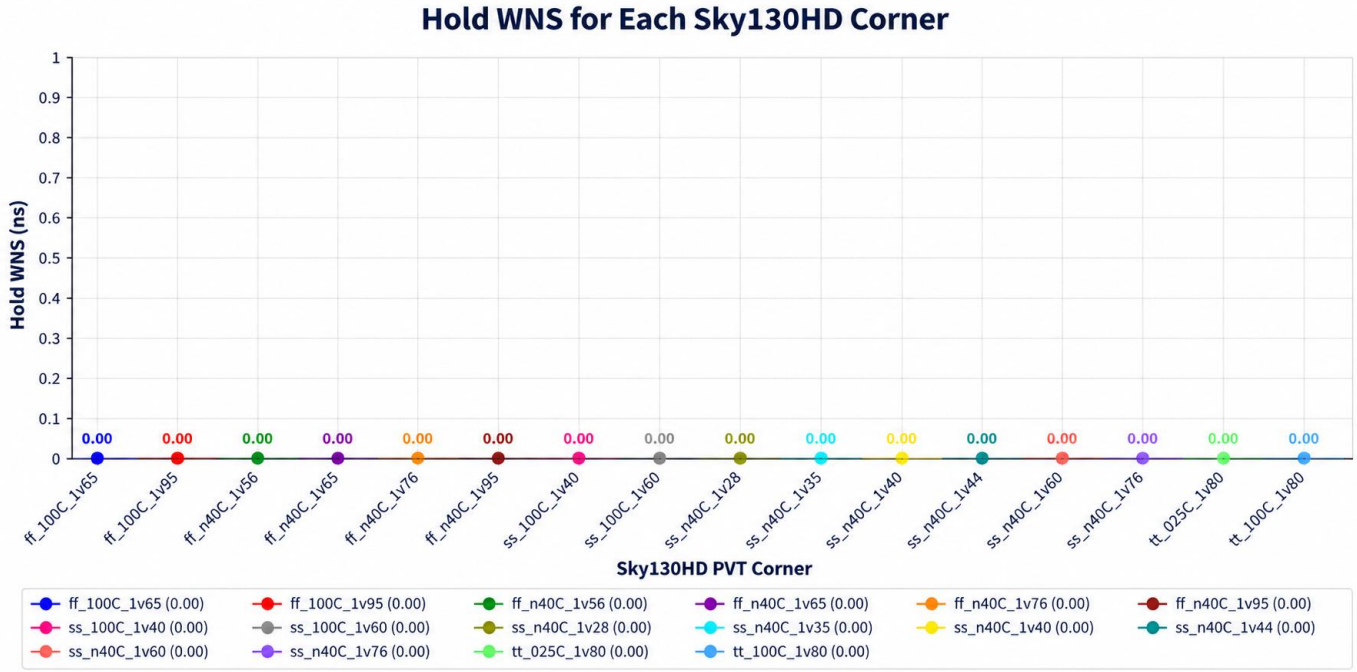
APPENDIX C:SETUP TNS



APPENDIX D:HOLD WS



APPENDIX E:HOLD WNS



APPENDIX F:HOLD TNS

